Presentation Script — Scenario & System Flow

1. Introduction

Good morning/afternoon everyone.
My name is **Simphiwe Nxumalo**, and without wasting any time, I've prepared a short **system workflow demonstration** to explain how everything works in order.

---

2. Overview of the System

The system starts with a **new employee** who needs to interact with the asset tracking platform.
In this system, there are **three main roles**:

1. **Staff Member** – responsible for managing assets, categories, and locations such as storage rooms.
2. **Verification Agent** – ensures data integrity by verifying that all asset information is correctly captured and stored.
3. **Administrator** – manages both users and assets, controls access, and monitors all activities.

If you look at the screen, you'll notice I have **two tabs open**:

- On the left: my **backend application**, built using **ASP.NET Web API**, which handles all the logic and data operations.
- On the right: my **frontend web application**, built using **ASP.NET MVC**, which provides the user interface.

---

3. Exploring the Web Frontend

Let's start by focusing on the **web front end**.
On the **Home Page**, you'll see a **clean landing page** that displays basic information and includes animations to enhance the user experience.

At the bottom of the page, there's a **button to download the Android APK file**.
This allows users to **download and install** the mobile version of the app.
Currently, you'll notice that downloading works fine, but due to Visual Studio's local environment restrictions, running certain upload/download features may appear limited — this works perfectly once deployed to a live environment.

We also have:

- An **About Page**, which provides details about me and the system background.
- A **Contact Page**, where users can find my contact details.

On the **top-right corner**, there are two key buttons:
Sign Up and **Login** — both supported by the **web** and **Android** versions.

---

4. Requesting Access (Sign Up Flow)

When a user clicks **Sign Up**, a **Request Access Form** appears.
Here, a new employee fills in personal details and uploads a verification document.

For demonstration, I'll skip the file upload since Visual Studio sometimes throws a visual rendering bug.

Once the form is submitted, the system sends a **notification to the Administrator** indicating a new access request has been received.

---

5. Administrator Actions (Access Management)

Now, I'll log in as an **Administrator** using one of my pre-configured accounts.

The **Android App** is restricted to **Staff Members**, so only administrators and verification agents use the web portal for management tasks.

After entering valid login credentials, if the **API is running** and the **connection string** is valid, I'm redirected to the **Administrator Dashboard**.
A welcome message appears via **SweetAlert**, confirming successful login.

On the dashboard, I can navigate to **"Manage Access"** — this page lists all users who have applied for system access.

From here, the admin can **Approve** or **Decline** requests.
For instance, by clicking the **Green Approve Button**, the system automatically updates the status and sends an **email notification** to the applicant confirming that their access has been approved.

---

6. Creating and Managing Users

After approving access, the administrator can also **create user accounts manually**, assigning appropriate roles such as Staff Member, Verification Agent, or Administrator.

Once a Staff Member account is created, that user can **log in using the Android app**.

---

7. Android App Integration

On the **Android App**, Staff Members can:

- **Log in** using their approved credentials.
- **Create new assets** by entering asset details.
- **Scan existing assets** using the **ZXing Library** for QR scanning.
- **Upload or generate QR codes** for assets.

If a new asset is submitted to the **API** without an existing QR code, the **backend automatically generates one** using the **QRCoder library**, ensuring that every asset in the system can be tracked digitally.

---

8. Closing

In summary, this system demonstrates a **complete asset management workflow** —
from **access request and approval**, to **asset registration, verification, and QR-based tracking** —
integrated seamlessly across **Web and Android platforms** using **ASP.NET Web API** as the backend.

Thank you for your time — I'll now proceed to the **live demonstration** of the workflow.